CURRICULUM OVERVIEW

Python Foundations

Build 9 Real Python Projects

A complete, hands-on journey into text-based Python. Students start from the very first line of code, then learn to solve problems, structure and debug programs, work with files and external libraries, and explore the tools used in modern Python development.



9 REAL PYTHON PROJECTS
From a calculator to live data

4 TESTS ALONG THE WAY
3 course tests + 1 final test

3	24	28	12–18
Courses	Lessons	Sessions in total	Age range

WHO IT IS FOR
Beginner friendly — no prior coding experience needed

BEFORE YOU START
No coding experience needed

LEARNING PROGRESSION

01 Python Programming Essentials
Write independent Python programs from the very first line.

02 Problem Solving & Program Design
Plan, debug, handle errors and build with files and classes.

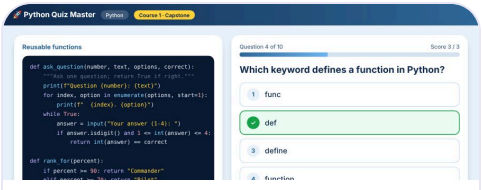
03 Python Tools & Data Foundations
Use real tools and data: notebooks, NumPy, Pandas and APIs.

Learn to think like a programmer
Students break down problems, design solutions and approach unfamiliar challenges independently, not just copy code.

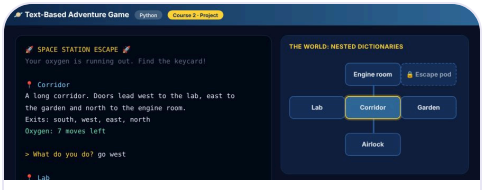
A portfolio, not just a certificate
Students finish with nine real projects they can build, demonstrate and add to their portfolios.

Tested, so progress is real
Every course ends with a test, and a final test closes the program, so progress is proven, not assumed.

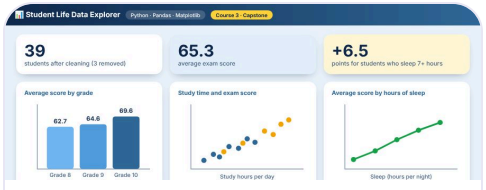
SAMPLE HANDS-ON PROJECTS STUDENTS BUILD



01 Python Quiz Master
Python · Strings · Collections



02 Text-Based Adventure Game
Python · Dictionaries



03 Student Life Data Explorer
Python · Pandas · Matplotlib

COURSE 01 · LESSONS 1–8

01 Python Programming Essentials

Write your first real Python programs, from scratch

Python 3 | Python editor | Control flow | Collections | Functions

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

No coding experience needed. Students learn how to write, understand and run Python programs, and gradually combine variables, decisions, loops, strings, collections and functions into interactive applications.

PORTFOLIO PROJECTS

- Smart Calculator
- Number Guessing Game
- Python Quiz Master

WHY IT MATTERS

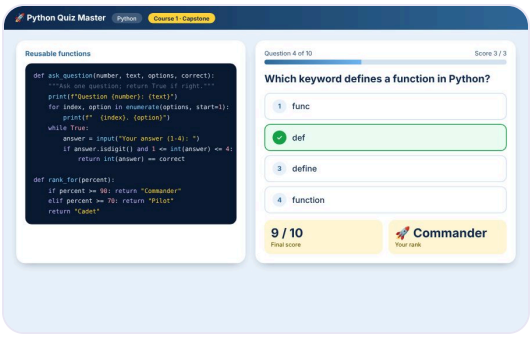
Every programmer starts here. Fluency with syntax, data types, conditions, loops and functions is the foundation for everything that follows, from games and websites to data science and AI.

8 LESSONS | **3** PROJECTS | **1** COURSE TEST

LESSON BREAKDOWN

1. Python Setup and Your First Program
2. Variables, Data Types, and Values
3. User Input, Output, and Operators
4. Making Decisions with Conditional Logic
5. Repetition with Python Loops
6. Working with Strings
7. Working with Python Collections
8. Creating Reusable Code with Functions

8 LESSONS — TEST UNLOCKS



SAMPLE HANDS-ON PROJECT

Python Quiz Master

Python · Strings · Collections

A quiz application built with strings and collections: it asks questions, checks the answers, keeps score and awards a rank.

COURSE TEST

Python Programming Essentials — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

COURSE 02 · LESSONS 9–16

02 Problem Solving & Program Design

Think, solve, structure and build like a programmer

Python 3 | Algorithms | Debugging | Files · JSON · CSV | OOP

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Students move beyond single statements and learn to approach problems systematically: designing solutions, organising larger programs, finding and fixing errors, handling the unexpected and working with persistent data.

PORTFOLIO PROJECTS

- Text-Based Adventure Game
- Library Management System
- Personal Expense Tracker

WHY IT MATTERS

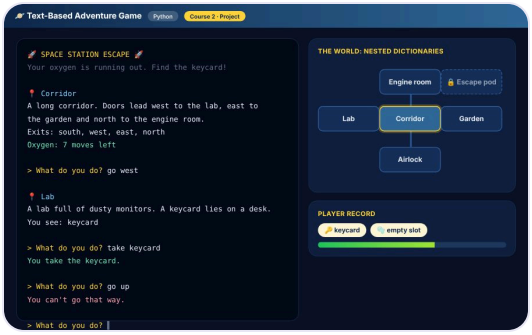
Writing code is only half of programming. Planning, testing, debugging and organising code into functions and classes is what turns a script into software that other people can trust.

8	3	1
LESSONS	PROJECTS	COURSE TEST

LESSON BREAKDOWN

1. Computational Thinking & Problem Decomposition
2. Functions in Depth & Modular Programming
3. Advanced Collections & Nested Data Structures
4. Algorithms & Problem-Solving Patterns
5. Debugging, Testing & Fixing Problems
6. Error Handling & Input Validation
7. Working with Files & Structured Data
8. Introduction to Object-Oriented Programming

8 LESSONS → TEST UNLOCKS



SAMPLE HANDS-ON PROJECT

Text-Based Adventure Game

Python · Dictionaries

A text adventure: the world is a nested dictionary, the player is a record, and small functions run the game.

COURSE TEST

Problem Solving & Program Design — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

COURSE 03 · LESSONS 17–24

03 Python Tools & Data Foundations

Prepare for real-world Python development

VS Code · Git · Jupyter · NumPy · Pandas · Matplotlib

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Students explore the tools professionals use: development environments, libraries and packages, Jupyter Notebooks, NumPy, Pandas and APIs. They take their first steps with real data and learn to document and share a finished project.

PORTFOLIO PROJECTS

- Student Life Data Explorer
- Dataset Explorer
- Live Data Explorer

WHY IT MATTERS

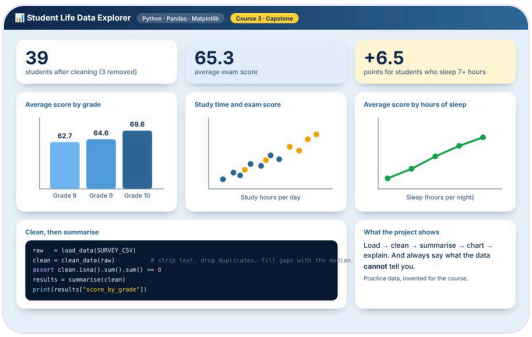
This course is the bridge to specialisation. It introduces the workflows and libraries behind data science and AI, so students arrive at those programs with confidence in their tools.

8 LESSONS | **3** PROJECTS | **1** COURSE TEST

LESSON BREAKDOWN

1. Python Development Environment & Workflow
2. Python Modules, Libraries & Packages
3. Introduction to Jupyter Notebooks
4. Introduction to NumPy
5. Introduction to Pandas & DataFrames
6. Data Cleaning, Summaries & Basic Visualization
7. Working with APIs & JSON Data
8. Python Capstone: Build, Document & Share

8 LESSONS → TEST UNLOCKS



SAMPLE HANDS-ON PROJECT

Student Life Data Explorer

Python · Pandas · Matplotlib

A data project that grows through the course: a survey summarised with Python's own modules and a clear notebook report, then upgraded with Pandas and charts.

COURSE TEST

Python Tools & Data Foundations — Course Test

Checks what you built and learned in this course. Taken after lesson 8, before the next course begins.

30 multiple-choice · 10 true / false · 40 marks

HOW STUDENTS ARE TESTED · 4 TESTS

Level-Up Checkpoints

Every course ends with a test of both ideas and practical skills, and a final test closes the program with questions and an independent Python project.



THE CHECKPOINT PATH — EVERY COURSE ENDS WITH A TEST, THEN ONE FINAL



COURSE TEST · every 8 lessons

Show what you built and learned

After every 8 lessons a course test checks the ideas and skills of that course, so each next course starts on solid ground.

30	multiple-choice questions — choose one of four answers
10	true / false questions
40	marks in total — 1 mark for every question

FINAL TEST · the finish line

One final test on the whole curriculum

One final test across the whole program: questions on Python, problem solving and tools, plus a practical programming task.

25	multiple-choice questions
5	true / false questions
20	marks for one small practical project (about 15 minutes) that your teacher marks

A TASTE OF A TEST

SAMPLE · MULTIPLE-CHOICE

Which Python function shows text on the screen?

- A input()
- B print() ✓ CORRECT
- C range()
- D len()

SAMPLE · TRUE / FALSE

A variable created inside a function can be used anywhere in the program.

True | False ✓

Examples in the style of a real test — written for this document, not taken from one.

AFTER EVERY TEST

Saved as you go
Answers save automatically while a student works, so nothing is lost if the connection drops.

Checked by the teacher
The teacher reviews the report and shares the result with the student.

Fair for everyone
Questions and answer options appear in a different order for every student.

The Complete Journey

Across 3 courses, 24 lessons and 4 tests, students progress from writing their first Python statement to building structured applications and exploring real-world data, finishing with nine projects for their portfolio.

3 COURSES · 24 LESSONS

01 Python Programming Essentials
Smart Calculator · Number Guessing Game · Python Quiz Master
8 lessons · + course test

02 Problem Solving & Program Design
Text-Based Adventure Game · Library Management System · Personal Expense Tracker
8 lessons · + course test

03 Python Tools & Data Foundations
Student Life Data Explorer · Dataset Explorer · Live Data Explorer
8 lessons · + course test

HOW STUDENTS ARE TESTED · 4 TESTS

COURSE TEST 1
Python Programming Essentials

COURSE TEST 2
Problem Solving & Program Design

COURSE TEST 3
Python Tools & Data Foundations

FINAL TEST
One final test on the whole curriculum
One final test across the whole program: questions on Python, problem solving and tools, plus a practical programming task.

BY THE END, STUDENTS CAN

- Write independent Python programs using variables, data structures, functions, conditions, loops and objects.
- Break down problems, design algorithms, debug, handle errors and work with files to build complete applications.
- Work with real Python tools and data: libraries, Jupyter Notebooks, NumPy, Pandas, APIs and basic charts.

Ready to start building?
All lessons and projects are taught live by an expert Lumexa instructor.

Lumexa Python Foundations · Ages 12–18